CP-FTMW spectrometer — bench note

Why the DUC carrier must sit on the k/T grid

Carrier phase coherence, coherent averaging, and the one condition that decides whether averaging helps or destroys the FID.

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One condition, and everything follows from it: f_NCO x T must be a whole number, where T is the buffer period. On grid, every shot is identical. Off grid, the carrier phase walks and averaging cancels the signal.

1. The signal model

In DUC mode the instrument plays a stored complex baseband against a hardware NCO:

$$s(t) \;=\; \mathrm{Re}\{\, z(t)\, e^{\,j(2\pi f_{NCO}\, t \,+\, \phi_0)} \},\qquad z(t) = I(t) + jQ(t)$$

Two very different kinds of behaviour hide in that one line.

$z(t)$ lives in segment memory and loops with period $T = n_{total}/rate_{cx}$. It satisfies $z(t + kT) = z(t)$ exactly — the same samples, the same DAC codes, on every repeat, forever.

The NCO is a free-running phase accumulator. It is not reset at the segment wrap and not reset by a trigger. Its phase simply keeps counting, $\phi(t) = 2\pi f_{NCO} t + \phi_0$.

2. What the scope captures on shot k

The marker fires at the same sample of the buffer every period, so shot k begins at $t_k = t_0 + kT$. Writing the captured record in local time τ (time since the trigger) and using $z(t_k + \tau) = z(\tau)$:

$$s_k(\tau) \;=\; \mathrm{Re}\{\, z(\tau)\, e^{\,j2\pi f_{NCO}\tau}\, e^{\,j\theta_k} \},\qquad \theta_k = \theta_0 + 2\pi f_{NCO} T\, k$$

That is the whole problem in one line. Every shot is the identical complex waveform multiplied by a scalar phasor. The marker gives a rigid TIME reference; it gives no PHASE reference, because the carrier never restarted with the buffer.

The per-shot phase step is therefore

$$\Delta\theta = 2\pi\varepsilon,\qquad \varepsilon \;\equiv\; \mathrm{frac}(f_{NCO}T) = \delta f \cdot T$$

where δf is how far the carrier sits from the nearest grid point k/T. If $f_{NCO}T$ is an integer then $\varepsilon = 0$, $\theta_k = \theta_0$ for every k, and all shots are bit-for-bit identical.

3. What averaging does to it

Averaging N shots sums those phasors — a geometric series, i.e. the Dirichlet kernel:

$$\frac{1}{N}\sum_{k=0}^{N-1} e^{j2\pi\varepsilon k} = e^{j\pi\varepsilon(N-1)}\frac{\sin(\pi\varepsilon N)}{N\sin(\pi\varepsilon)} \quad\Longrightarrow\quad |G(N,\varepsilon)| = \frac{|\sin(\pi\varepsilon N)|}{N|\sin(\pi\varepsilon)|}$$

Noise is uncorrelated shot to shot, so its amplitude still falls as $\sqrt{N}$ while the signal is multiplied by $|G|$. Hence:

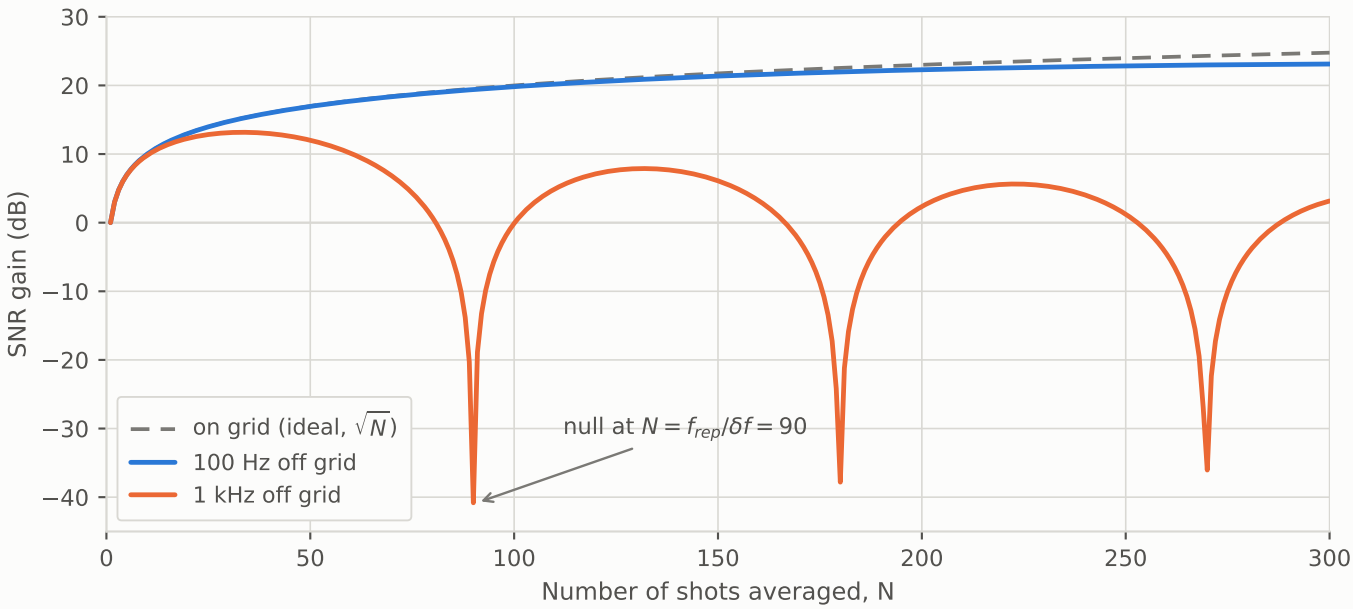
$$\text{SNR gain} = 10\log_{10} N + 20\log_{10}|G|, \qquad N_{eff} = N|G|^2$$

$|G| = 1$ only at $\varepsilon = 0$. For small ε it degenerates to $|G| \approx \text{sinc}(\varepsilon N)$: the loss depends on the phase walk accumulated over the WHOLE average, not on ε alone. That is why the requirement gets harsher the more you average.

4. Numbers for the default shot

n_{total} = 12,512 complex samples, so T = 11.1218 us and the grid spacing is $f_{rep} = 1/T$ = 89.914 kHz. Coherent gain $20\log_{10}|G|$, in dB:

delta-f (Hz)	epsilon	N=2	N=8	N=64	N=1000
45,000	0.500480	-56.4	-56.4	-56.4	-60.0
10,000	0.111218	-0.5	-18.1	-35.6	-54.7
1,000	0.011122	-0.0	-0.1	-9.1	-39.4
100	0.001112	-0.0	-0.0	-0.1	-20.1
1	0.000011	-0.0	-0.0	-0.0	-0.0



SNR gain against the number of shots averaged, for two carrier offsets from the nearest grid point (T = 11.122 us). Larger offsets null even sooner — 10 kHz at N = 9 — and are left to the table above. The 1 kHz curve is the one to look at twice: it peaks near N = 64 and then collapses. Averaging harder is not a remedy.

Three things to take from it.

Half a grid step off (45 kHz) is total loss at any N. Successive shots are ~180 degrees apart, so two of them already cancel.

Averaging is non-monotonic off grid. More shots can make the FID SMALLER — on the bench that looks exactly like a hardware fault, which is why it is worth being able to name.

The budget tightens as 1/N. For 0.5 dB of loss or less you need $\varepsilon N \le 0.186$, i.e. $\delta f \le 0.186\, f_{rep}/N$.

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delta-f = 1 kHz, the practical case:

   N =   16    ideal  12.0 dB      actual   11.6 dB
   N =   64    ideal  18.1 dB      actual    9.0 dB
   N =   90    ideal  19.5 dB      actual  -40.8 dB   <- null at f_rep/delta-f
   N =  256    ideal  24.1 dB      actual   -1.7 dB

-0.5 dB budget   delta-f <= 0.186 * f_rep / N:

   N =     64   ->  delta-f <=   261.3 Hz
   N =   1000   ->  delta-f <=    16.7 Hz
   N =  10000   ->  delta-f <=     1.7 Hz
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> **At 1000 averages the carrier must be within ~17 Hz of a 90 kHz grid point. 'Close enough' is not a strategy — you snap.**

5. Why snapping is exact, and one detail in the driver

Set the carrier to an exact multiple of the repetition rate:

$$f_{NCO} = k\frac{rate_{cx}}{n_{total}} \qquad \Longrightarrow \qquad f_{NCO}\,T = k\frac{rate_{cx}}{n_{total}} \cdot \frac{n_{total}}{rate_{cx}} = k \quad \text{exactly}$$

The sample rate cancels. The coherence condition is a RATIO OF INTEGERS, so it stays satisfied even if the 9 GS/s clock is off by ppm or drifts with temperature. That is why this belongs in software rather than in a hunt for clock accuracy.

One caveat from our own driver: _duc_upload_ch writes ':NCO:CFR1 {ncof:.0f}' — rounded to 1 Hz. At n_{total} = 12,512 the grid is 89,913.6829 Hz, not an integer, so the write leaves up to 0.5 Hz of residual. Harmless at realistic N (-0.5 dB only around N = 33,000, first null near N = 180,000), but it does cap how far averaging can go.

It is free to remove. Choose n_{total} so that it DIVIDES $rate_{cx}$ = 1.125e9 exactly: the grid then falls on a whole number of Hz and the 1 Hz write is exact. The multiples of 32 that qualify are

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  64, 96, 160, 192, 288, 320, 480, 576, 800, 960, 1,440, 1,600, 2,400,
  2,880, 4,000, 4,800, 7,200, 8,000, 12,000, 14,400, 20,000, 24,000, 36,000, 40,000, 60,000

  smallest that fits the 1 / 0.1 / 10 us shot (needs >= 12,487):  14,400
    ->  T = 12.8000 us,   grid = 78,125 Hz exactly,   detect = 11.70044 us
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6. Absorbing the remainder — exact frequency AND coherence

Snapping the carrier does not have to move the output frequency. Snap f_{NCO} to the nearest k/T, then put the leftover into the BASEBAND:

$$z'(\tau) = z(\tau)\, e^{j2\pi\delta\tau}, \qquad \delta = f_{RF} - f'_{NCO}$$

The output stays centred at $f'_{NCO} + \delta = f_{RF}$, while the per-shot phasor becomes $e^{j2\pi f'_{NCO} Tk} = e^{j2\pi k} = 1$ for every k. It works because the baseband is memory-played: whatever δ is baked into it replays identically every period by construction.

This is also where the two different 'snap to the grid' rules meet, and they must not be confused:

> GATED signals (the CP-FTMW chirp and its LO) are windowed to zero at both ends, so they never reach the buffer wrap and δ can be anything. Result: any exact RF frequency, fully coherent.
>
> CONTINUOUS signals must additionally hold a whole number of cycles in T, or the phase steps at the wrap and sprays spurs — the rule the waveform builder already enforces (AWG.snap_continuous). Both conditions together force $f_{RF}T$ to be an integer, so a continuous DUC CW is restricted to the grid.

Gated: exact frequency and full coherence. Continuous: pick from the 78-90 kHz grid. The CP-FTMW shot is gated, so it loses nothing.

7. What this does not cover, and what is still unproven

The derivation assumes the trigger is rigid relative to the buffer. That is true for an AWG marker, and false for an external pulse: the P908xD external trigger jitters by 32 SCLK periods (~3.6 ns at 9 GS/s), which is many carrier cycles. Chain the timing (master pulse -> AWG TRIG 1 -> AWG marker -> scope); never fan it out.

It also assumes the scope timebase is referenced to the AWG. Independent oscillators drift the sampled time axis from shot to shot — a separate decoherence mechanism, which is what set_reference_external() exists to remove.

And the premise itself is inferred, not proven: that the Proteus NCO genuinely free-runs across the wrap. The -176 to +155 degree per-acquisition phase scatter observed earlier is consistent with it, which is not the same as evidence for it.

The clean test: one shot on grid, one deliberately off grid, N captures on the marker trigger, per-shot carrier phase plotted. Off grid the phase should walk by exactly $2\pi\varepsilon$ per shot — and the measured walk gives ε directly.

Status: the waveform builder implements the continuous-signal rule (AWG.snap_continuous, previewed in the GUI). NCO snapping is NOT implemented — it is deliberately parked with the marker-trigger work, since neither is useful without the other.

Numbers in this note were computed, not quoted; see tests/test_awg_builder.py for the builder's own checks.